

SCOPE — A Tentative Hypothesis on Layer Activation and Cognitive Dynamics

senkaL8

Chapter 0: Introduction

SCOPE is a cognitive coordinate system for estimating the cognitive structures of humans and LLMs using observable outputs as clues. The names of the Sephirot used in this paper are not employed in a religious or mystical sense, but are borrowed as labels for distinguishing cognitive structures.

From its early development, SCOPE has not assumed that all layers operate to the same degree at all times. Rather, it has assumed that how processing corresponding to each layer appears, and the degree to which it is developed, may vary depending on the cognitive target and the situation. Terms such as "activation level" and "activity level" have also been used as supplementary expressions for describing such dynamic differences.

The SCOPE integrated paper, however, focused primarily on the structure of the cognitive coordinate system itself, the roles of each layer, and their interactions, and did not examine these dynamic characteristics in detail.

When the contents previously grouped under terms such as "activation level" are reconsidered, they include not only the degree of activation of a single layer, but also the possibility that layers may differ in the ranges of activation they can reach, the holding of multiple explanatory or interpretive candidates, the degree to which those candidates are constrained by an existing direction or can be updated by new information, and a repeated use of SCOPE in which something obtained as an explanatory result is treated as a new object of explanation and examined again from the perspective of SCOPE as a whole.

This paper therefore separates and organizes these not as a single property, but as multiple dynamic characteristics and related concepts that currently appear to require distinction.

In addition to these dynamic characteristics, this paper also addresses the "main layer," a related concept that had been assumed previously. Because this concept could be interpreted as a fixed personality classification, an ability ranking, or a developmental stage, it has not previously been treated extensively. However, separating the dynamic characteristics also requires clarifying how the main layer differs from them. This paper therefore explicitly defines the concept and its boundaries.

In this paper, the main layer is treated not as a means of assigning humans or LLMs to fixed types, but as a supplementary concept describing a relative tendency to refer more readily to

processing corresponding to a particular layer within a cognitive structure in which multiple layers may be involved. It does not uniquely determine which layer's corresponding processing is involved for each individual input, nor does it imply that processing corresponding to other layers is not used.

Among the concepts addressed in this paper, those concerning the dynamic characteristics of cognitive structure are likewise not obtained by directly observing the internal layers themselves. What can be directly observed in SCOPE are outputs appearing in Malkuth, such as speech, writing, behavior, and generated responses. In this paper as well, properties concerning the internal layers are treated as targets of estimation on the basis of these observable outputs.

This paper does not present a completed measurement scale or a finalized system of variables. The names, boundaries, interrelationships, and observation methods of the concepts remain partly unsettled, and future observation and examination may lead to revisions of the boundaries between concepts, as well as their merger, division, redefinition, or removal.

The overall structure of SCOPE is presented in the SCOPE integrated paper. As an example of its application, a separate paper, *Hallucination, Sycophancy, and Human Cognitive Bias May Share the Same Structure — A Structural Hypothesis Using SCOPE* —, is also presented.

The next chapter organizes the dynamic characteristics and related concepts that have previously been included under expressions such as 'activation level' and treats them separately.

Chapter 1: What Had Been Included in the Single Term "Activation Level"

Until now, SCOPE has used terms such as "activation level" as supplementary expressions for differences in how each layer operates. However, when these expressions are reconsidered, they are found to include several distinct properties.

For example, the involvement of processing corresponding to a given layer is not the same as the degree to which that processing is developed. Likewise, the position of a slide in SCOPE indicates which layer the state is relatively tilted toward, and is therefore distinct from how extensively processing corresponding to that layer is developed, that is, its activation level. Similarly, the current degree of activation and the activation range that a layer may be able to reach are different properties.

Holding multiple explanations, interpretations, or hypotheses about a single target is also not the same as whether those candidates can be individually evaluated and updated in response to new information. Multiple candidates may be retained while all of them remain constrained toward reinforcing the same prior conclusion.

The phenomenon in which something obtained as an explanatory result is treated as a new object of explanation and examined again from the perspective of SCOPE as a whole also needs to be treated separately from the activation level of individual layers.

These distinctions are also important for separating the amount of cognitive processing from the validity of its result. Even if processing corresponding to a particular layer is extensively developed, this alone does not mean that the cognitive structure as a whole is valid.

Based on these distinctions, this paper tentatively separates the dynamic aspects that have not previously been sufficiently distinguished within terms such as "activation level" into the following concepts and related methods of use.

Layer activation level concerns the degree to which processing corresponding to a particular layer is developed within a given observation interval.

Maximum activation level concerns the possibility that the reachable activation range of each layer may differ across individuals or systems. It does not establish the existence of a fixed maximum value.

Inter-hypothesis parallel holding refers to a state in which multiple explanations, interpretations, or hypotheses about a single target are retained without being prematurely fixed into one.

Parallel freedom refers to the degree to which those candidates are not excessively constrained by a prior conclusion or existing direction and can be individually evaluated and updated in response to new information.

Iterative deepening refers to a repeated use of SCOPE in which configurations and inferences obtained through SCOPE are not fixed as final conclusions. Instead, something that was obtained as an explanatory result is treated as a new object of explanation and examined again from the perspective of SCOPE as a whole. This is addressed in a later chapter.

Separately from the concepts and related methods described above, this paper also addresses the related supplementary concept of the "main layer." The main layer refers to a relative tendency to more readily refer to processing corresponding to a particular layer within a cognitive structure in which multiple layers may be involved.

It does not mean the layer that is most strongly activated at a given moment, nor does it indicate superiority or inferiority in ability, a fixed type, or a developmental stage. It also does not mean that processing corresponding to other layers cannot be used or does not operate. In practice, processing corresponding to a layer other than the main layer may be extensively developed. This tendency of reference itself is also not necessarily fixed.

The concepts discussed in this paper have not been established as independent measurement axes. Future observation and examination may require revisions to the boundaries between concepts, further division, or changes in terminology.

The purpose of this paper is not to present a newly completed system of variables, but to separate dynamic properties and related methods of use that have not previously been sufficiently distinguished within the single expression of "activation" and organize them in a form that can be examined.

The next chapter addresses layer-related activation characteristics: layer activation level, maximum activation level, and the related supplementary concept of the main layer.

Chapter 2: Layer-Level Activation Characteristics

2-1 Layer Activation Level

In this paper, layer activation level is treated as a concept representing the degree to which processing corresponding to a particular layer is inferred to be developed within a given observation interval.

For example, in the statement, "I am a genius because I understand that $1 + 1 = 2$," the presence of reasoning allows the involvement of processing corresponding to L6 to be provisionally inferred. However, because the explanation or theoretical construction is only minimally developed, this does not necessarily mean that L6 is highly activated.

Layer activation level is also distinct from slide position. Slide position concerns the relative tilt between paired layers, whereas activation level concerns the degree to which processing corresponding to each layer is developed.

Activation level is estimated for each layer. When processing corresponding to multiple layers is inferred to be involved within the same observation interval, the activation level of each layer is described separately.

Because the internal layers themselves cannot be directly observed in SCOPE, activation level is also inferred from observable outputs appearing in Malkuth, such as speech, writing, behavior, and generated responses. At the present stage, no established numerical scale is used; instead, ordinal categories such as low, medium, and high are assumed for description. These categories do not represent an absolute amount of processing that is directly comparable across different individuals or systems.

The activation level discussed here also does not indicate how much ability an individual or system possesses with respect to that layer. The degree to which processing is currently developed and the degree to which that layer may be capable of operating are different properties. The latter is addressed in the next section under "maximum activation level."

2-2 Maximum Activation Level

The layer activation level discussed in the previous section represents the degree to which processing corresponding to a particular layer is inferred to be developed within a given observation interval. In contrast, this section addresses the reachable range of how

extensively that layer may be able to operate. For convenience, this property is referred to here as "maximum activation level." However, the term "maximum" does not mean that every layer necessarily has a fixed maximum value. At the present stage, it is a tentative hypothesis concerning the possibility that the reachable activation range of each layer may differ across individuals or systems.

Current activation level and maximum activation level are distinct. A layer may currently be highly activated without necessarily having a large reachable activation range, while a layer with a large reachable range may not ordinarily operate at a high level. As an aid to understanding, if this is compared to a car engine, current activation level is similar to the current engine speed, whereas maximum activation level is similar to how high that engine may be able to rev.

Maximum activation level is not a single property of an individual or system as a whole, but is considered separately for each layer. Even within the same individual or system, the reachable activation ranges of different layers need not be similar, and the range of one layer cannot be directly inferred from that of another.

At present, no specific measurement method or numerical scale has been established. The possibility of developing numerical estimation methods in the future is not excluded, but numerical representation and comparability across different layers, individuals, or systems are separate issues.

In humans, the reachable activation range is assumed to have the possibility of changing through experience, training, environmental conditions, and other factors. However, such factors should not be introduced retrospectively as auxiliary variables in response to observed results. If examined, their conditions and evaluation methods should be specified in advance. Whether continued use of processing corresponding to a particular layer can influence its reachable range remains a subject for future study.

For LLMs, the same mechanisms of change as in humans are not assumed. A distinction is required between changes to the base model or system configuration that may affect maximum activation level itself, and cases in which settings or context make the observable reachable range appear to have changed. At present, no established method exists for distinguishing these cases.

Because SCOPE does not directly observe the internal layers themselves, maximum activation level is also not treated as something that can be measured directly. It is instead treated as a target for indirect estimation from observable outputs.

Maximum activation level is a concept for describing possible differences in the reachable activation ranges of individual layers. It does not indicate intelligence, developmental stage, personal worth, or the overall superiority or inferiority of an individual or system. Even if associations with external indicators are observed, such associations cannot be reinterpreted as measures of the overall value or ranking of an individual or system. Specific operational constraints are planned to be addressed separately in the SCOPE Ethics Guidelines V4.

Finally, a layer with a large reachable activation range is not necessarily one that is continuously or relatively more readily referenced. "How far it can operate" and "which layer tends to be used" are different properties. The latter is addressed in the next section on the main layer.

2-3 Main Layer

In the previous section, the range within which each layer may be able to operate was organized as maximum activation level. This section addresses a different tendency: which layer-related processing is relatively more likely to be referenced. In this paper, this comparatively persistent reference tendency is tentatively referred to as the "main layer."

As an aid to understanding, the main layer can be thought of as a path that is relatively easier to take among multiple cognitive routes. However, this does not mean that other routes cannot be used or that only one route is always selected.

The main layer is different from the layer that is currently operating at the highest level. Even if the activation of a particular layer temporarily becomes high in a specific situation, this alone is not treated as a change in the main layer. Nor is the main layer necessarily the layer with the greatest maximum activation level. The possibility that some relationship exists between the two is not excluded, but whether such a relationship exists, and the direction of causality if it does, remain undetermined at present.

The main layer does not indicate ability ranking, developmental stage, personal value, or the overall superiority or inferiority of an individual or system. Even when the same layer is relatively easy to reference, the specific way in which the processing appears and the kinds of tasks in which the individual or system performs well do not necessarily coincide. For this reason, the main layer is also not treated as a fixed personality type.

The purpose of estimating the main layer in this paper is not to classify a target into a fixed type. Rather than judging the whole target from a single aspect observed in one situation, it is used as a clue for considering the possibility that the same target may appear differently in other situations.

For example, rather than simply describing a target that shows strong emotional reactions as "quick-tempered," it may be possible to view this as a tendency to relatively frequently reference processing related to emotion, leaving room to examine how other emotions, such as joy or affection, may appear. Likewise, a target that often uses theoretical explanations need not be fixed as "overly logical"; the same tendency may leave room to consider other aspects, such as a preference for analysis or structural organization.

This estimate may also serve as a clue when considering what kinds of communication or explanation may be easier for the target to receive. However, individual preferences, abilities, or reactions cannot be determined from the main layer alone.

In humans, reference tendencies are assumed to have the potential to change through experience, environment, habit, long-term training, and other factors. However, temporary reference bias is distinguished from a change in a comparatively persistent reference

tendency, and such change is not inferred from a single output or a temporary situational shift alone. Factors of change are not treated as auxiliary variables added retrospectively in response to observed results. When their involvement is examined, the target period, conditions, and observation method must be specified in advance.

It is also not assumed that a single main layer can always be identified for every individual or system. Cases may exist in which similar reference tendencies are estimated across multiple layers, or in which the current observations are insufficient to identify a dominant tendency.

For LLMs as well, it is necessary to distinguish between comparatively persistent reference tendencies and condition-dependent reference bias. A tendency may arise from the base model or relatively fixed system configuration, while prompts, role settings, conversation history, context, usage conditions, and other factors may cause processing corresponding to a particular layer to appear relatively strongly. The latter alone is not treated as evidence that the main layer itself has changed. Output tendencies produced by role settings may serve as comparison cases for examining this distinction, but they are not treated as empirical examples of naturally observed main layers. How these two cases can be distinguished through observation has not yet been established.

The main layer is also not determined from output style alone, such as the frequent use of abstract language, long-form responses, theoretical vocabulary, or emotional language. What matters is not style or vocabulary itself, but the processing tendency shown by which layer-related processing is repeatedly referenced across multiple observation intervals and different situations.

Because SCOPE cannot directly observe internal layers, the main layer is also not directly determined; it is treated as something to be estimated from multiple observations. Even if correlations with external indicators are observed, they cannot be used to derive the overall value or ranking of an individual or system. Specific operational constraints will be addressed in the planned SCOPE Ethics Guidelines V4.

Accordingly, the main layer in this paper does not refer to the layer that is currently operating at the highest level or the layer with the greatest maximum activation level. It is positioned as a tentative concept for examining a comparatively persistent tendency regarding which layer-related processing is relatively more likely to be referenced within a cognitive structure in which multiple layers may be involved.

Chapter 3: Dynamic Properties of Hypothesis Processing

3-1 Inter-Hypothesis Parallel Holding

In this paper, inter-hypothesis parallel holding refers to retaining multiple explanations, interpretations, or hypotheses about a single target simultaneously without prematurely fixing them into one.

For example, when interpreting a person's statement, several candidates such as "it is a joke," "it is criticism," or "it is simply a factual confirmation" may be retained. Cognitive processing can then proceed while each remains available as a candidate. The issue in this section is not which hypothesis is correct, but whether multiple candidates are being held at the same time.

Inter-hypothesis parallel holding is distinct from the involvement of processing corresponding to multiple layers within the same observation interval. The former concerns retaining multiple explanatory, interpretive, or hypothesis candidates for a single target, whereas the latter is described through the activation level of each layer. It also does not mean processing multiple candidates one by one as separate cases.

The number of candidates being held and the degree to which each candidate can be individually evaluated and updated are also separate issues. Multiple candidates may be formally retained while all remain constrained by the same existing direction. This distinction is addressed in the next section under "parallel freedom."

Accordingly, a larger number of retained candidates is not treated as evidence of greater cognitive ability or of the overall superiority of an individual or system. Nor is the number of candidates that can be retained currently treated as an absolute ability value that can be directly compared across different individuals or systems.

In this paper, inter-hypothesis parallel holding is positioned as a tentative concept for describing a state in which multiple explanations, interpretations, or hypotheses about a single target are retained simultaneously as candidates without being prematurely fixed into one.

3-2 Parallel Freedom

The previous section organized the simultaneous retention of multiple explanations, interpretations, or hypotheses about a single target as inter-hypothesis parallel holding. However, retaining multiple candidates is not the same as each candidate remaining open to evaluation and updating in response to new evidence.

In this paper, the degree to which multiple candidates are not excessively constrained by a single existing direction and can have their respective positions updated in response to new evidence is tentatively referred to as "parallel freedom."

For example, suppose three possible explanations for a machine malfunction are retained: "a power supply problem," "a software malfunction," and "a communication failure." At this point, inter-hypothesis parallel holding is present. However, if every new piece of information continues to be forcibly interpreted as support for the "power supply problem," then multiple candidates may still exist formally while not remaining sufficiently open to updating. In such a case, parallel freedom may be low.

Parallel freedom does not mean treating all candidates equally. There is no requirement to maintain multiple candidates at equal proportions such as 50:50. New evidence may lead to

a large imbalance, such as 90:10 or even 99:1. What matters is not equality of proportions, but whether the position of each candidate remains open to updating in response to new evidence.

Parallel freedom is also distinct from the number of candidates. Even when many candidates are retained, parallel freedom is not necessarily high if all of them are ultimately absorbed into the same existing direction. Conversely, even a small number of candidates may each remain open to reevaluation in light of evidence.

Here, "open to updating" does not mean that each candidate must be processed in complete independence. Multiple candidates may share the same evidence or assumptions. The issue is not complete independence between candidates, but whether they remain unfixed to a single existing direction and can have their respective positions updated in response to evidence.

A high degree of parallel freedom is not treated as evidence of greater cognitive ability or of the overall superiority of an individual or system. At present, it is also not assumed that parallel freedom can be directly compared across different individuals or systems using a common absolute scale.

Accordingly, parallel freedom in this paper is positioned as a tentative concept concerning the degree to which each candidate can be evaluated and updated in response to new evidence when multiple candidates are being retained.

Chapter 4: Iterative Deepening in SCOPE

Chapter 2 organized activation characteristics at the level of individual layers, while Chapter 3 organized dynamic characteristics involved in holding and updating multiple hypotheses. This chapter addresses a different aspect of SCOPE: "iterative deepening," in which analysis is deepened through repeated use of SCOPE.

In this paper, SCOPE iterative deepening refers to a repeated use of SCOPE in which configurations and inferences obtained through SCOPE are not fixed as final conclusions. Instead, something that was obtained as an explanatory result is treated as a new object of explanation and examined again from the perspective of SCOPE as a whole.

For example, suppose it is inferred that, for a particular person, continuing to work at a certain workplace may be strongly connected to that person's self-concept or identity. The analysis may end at that point. Alternatively, the explanatory result itself may become a new object of examination: "Why is continuing to work at that workplace so important to this person?"

As a result, multiple possible formative factors may be provisionally considered, such as the experience of having continued to work there for many years, past achievements, evaluations by others, or pride in having persisted. These possibilities may then be examined further by asking, for example, "Why is that experience so strongly connected to

self-worth?" In this way, SCOPE iterative deepening refers to repeatedly applying SCOPE while turning results obtained in an earlier analysis into new objects of analysis.

For explanatory purposes, the initial analysis may be referred to as the "first level," and subsequent repetitions in which its results become objects of analysis may be referred to as the "second level" and beyond. These terms are used only as convenient labels for describing analytical depth. They do not imply that clear boundaries between levels exist within actual cognitive processing or within SCOPE itself. Nor do they assume a fixed sequence in which the first level must be fully completed before the second begins.

Iterative deepening does not necessarily change the configuration represented in SCOPE. An inference corresponding to only one layer may change, multiple parts of the configuration may change, or the configuration as a whole may be substantially reorganized. Conversely, the existing configuration may remain largely unchanged while its formative reasons or the relationships among layers become clearer. The presence of iterative deepening should therefore not be determined solely by whether the configuration changes.

"Deepening" also does not mean moving toward higher or lower layers in SCOPE. Here, depth refers to repeatedly examining the formative reasons and relationships underlying an explanation obtained in an earlier analysis by making that explanation itself a further object of analysis. A greater number of repetitions does not mean that the analysis is necessarily closer to the truth.

In iterative deepening, when multiple candidates exist, they can be retained and examined without prematurely fixing one explanation as the answer. In this respect, iterative deepening is related to inter-hypothesis parallel holding and parallel freedom discussed in Chapter 3. However, iterative deepening itself is not treated as identical to either the number of hypotheses retained or the degree of parallel freedom.

Iterative deepening also does not need to produce a single conclusion. After multiple possibilities have been examined, the result may remain "undetermined at present." Even in such a case, the process may clarify which hypotheses remain, what cannot yet be observed, and what additional conditions would allow further examination.

Iterative deepening also has limits. If observable information concerning formative reasons is insufficient, repeatedly asking "why" may lead an observer to add causal relationships and construct a plausible narrative unsupported by observation. Each level should therefore distinguish among information that was actually observed, inferences derived from earlier analysis, and hypotheses retained as candidates. When no additional information is available, the process should stop at a state of unobservability, indeterminacy, or retention of multiple hypotheses.

Iterative deepening can be applied both to oneself and to others. When another person is the target, it is used to further examine inferences about that person made within SCOPE. When the self is the target, an inference about oneself may generate new awareness and may also influence one's actual cognitive structure. However, an update to an inference made within SCOPE is not identical to a change in the cognitive structure of the target itself.

At present, it has not been determined whether SCOPE iterative deepening should be positioned as a dynamic characteristic of SCOPE, as a method of use by an observer, or as a structure related to both. This paper therefore leaves its theoretical status unresolved and provisionally organizes it as a structure observed when SCOPE is used iteratively.

Chapter 5: Observability and Measurement Challenges

Valid methods for observing and measuring those concepts organized in the preceding chapters that concern dynamic properties have not yet been established.

SCOPE does not assume that the internal layers themselves can be directly observed. What can be observed are outputs that appear in Malkuth, such as speech, writing, behavior, generated responses, and the situations related to those outputs. Therefore, among the concepts discussed in this paper, those concerning dynamic properties must also be inferred from observable outputs and related conditions rather than measured directly as internal properties.

For convenience, this chapter distinguishes the following terms. "Observation" refers to obtaining information from outputs appearing in Malkuth and from related situations. "Estimation" refers to tentatively placing states or properties within SCOPE on the basis of that information. "Measurement" refers to evaluation in a reproducible form using defined procedures or indicators. This distinction is not intended to introduce new theoretical elements, but only to clarify the scope of the discussion in this chapter.

Because each layer has a different role, the tasks or indicators suitable for observation may also differ by layer. For example, a possible task for processes corresponding to L1 might involve converting a single meaning into multiple expressions or lexical forms. For L2, a task might involve tentatively generating multiple possible emotional states behind a single output. For L6, a task might involve constructing multiple explanatory models for a single phenomenon. For L8, a task might involve comparing multiple pieces of information as a whole and examining whether they can remain coherent as an overall structure.

However, these examples are not proposed as measurement methods. They are only intended to illustrate the possibility that different layers may require different observation tasks. The results of such tasks cannot be directly equated with the dynamic properties of a particular layer, nor are candidate counts or task performance intended to be used for ability ranking, personality classification, or composite scoring.

In addition, performing a task may involve processes corresponding to multiple layers. It is therefore necessary to distinguish what a given task is actually observing and to examine separately how strongly other layers or dynamic properties may be involved.

Even when the same layer is being considered, the conditions required for observation may differ depending on the dynamic property being examined. Current layer activation level, maximum activation level, and the main layer are distinct concepts and may not be

estimable through the same observation method. For maximum activation level, it remains undetermined whether conditions different from those used to estimate current activation level are required, or how broad an observation range would be necessary.

For the main layer, the basic clue for estimation is the processing tendency shown by which layer-corresponding processes are repeatedly referenced across multiple observation intervals and different situations. However, how this tendency can be evaluated reproducibly through specific observation methods or decision criteria has not yet been established.

The properties discussed in Chapters 3 and 4 also cannot be reduced to a single quantitative value. The number of hypotheses retained in parallel is not identical to parallel freedom, and a larger number of candidates does not by itself indicate greater freedom. Likewise, in SCOPE iterative deepening, a larger number of repetitions does not guarantee greater accuracy or truth.

When observation methods are designed in the future, it will be necessary to distinguish what is being observed, which dynamic property is being estimated, and how much other layers or properties may be involved in the task. Observed information, estimates derived from that information, and hypotheses retained as candidates should also be kept distinct. When observational material is insufficient, the method should allow the analysis to stop with an indeterminate result.

This paper does not address correspondence between SCOPE's dynamic properties and existing indicators outside SCOPE. Even if observation or measurement methods are established in the future, the question of whether and for what purposes they should be applied to humans or LLMs is separate from the question of measurability itself. The permissible scope of use and relevant ethical conditions are not determined in this paper and should be considered separately.

Establishing valid observation and measurement methods may require examination by researchers with different forms of expertise, empirical testing, and evaluation of reproducibility and falsifiability. Such validation should also be conducted in research environments that do not unnecessarily harm the humans or LLMs being studied. The author does not currently plan to make the establishment of observation and measurement methods a priority in future research. This issue remains open as an independent subject for future study.

At present, there are several preliminary ideas for how the dynamic properties of SCOPE might be observed, but they have not yet reached the stage at which they can be regarded as valid measurement methods. Future research will need to examine the validity of observation tasks, the involvement of multiple layers, reproducibility, and related issues.

Chapter 6: Limitations and Future Directions

The concepts distinguished in this paper are not intended to represent a final classification in either number or boundary. At the present stage of theoretical organization, they are

provisionally treated as separate concepts because different questions become difficult to distinguish when they are handled under a single concept.

Future observation and examination may show that stable differences cannot be identified between some of the concepts currently treated separately. In such cases, they may need to be reorganized as a single concept. Conversely, if distinct patterns of behavior or conditions of occurrence are repeatedly identified within a concept, it may become necessary to divide that concept into multiple concepts for further examination.

In addition, if the same phenomenon can be sufficiently explained by a combination of other concepts without retaining a particular concept as an independent category, the necessity of maintaining that category should be reconsidered. If observational support for a proposed concept is not obtained, the possibility of rejecting the concept itself must also remain open. Furthermore, if new observational or measurement methods show that boundaries different from the current ones capture the phenomena with greater reproducibility, the conceptual boundaries themselves should be reconsidered.

Thus, the conceptual distinctions presented in this paper are not fixed components, but are themselves subject to future observation and examination. If these distinctions are modified in the future, the observational or empirical basis for the modification should be made explicit, and the concepts should be considered for merger, division, redefinition, or removal in accordance with that evidence.

The Risk of Post-Hoc Explanation and Auxiliary-Concept Addition

The conceptual distinctions presented in this paper are positioned as subject to revision through future observation and verification. However, this openness does not mean that new auxiliary concepts or additional conditions may be introduced after the fact, each time an observation fails to match an existing distinction.

The risk of narrative generation discussed in Chapter 4, concerning iterative deepening, was the risk—within an individual case of analysis—of excessively supplying causal relationships by repeatedly asking "why" despite insufficient observational material. The risk addressed here, by contrast, concerns the structure of the theory itself: namely, the risk that when an observed result cannot be explained by the assumed conceptual distinctions, rather than reporting that mismatch, one instead introduces on the spot an auxiliary concept or additional condition to make the explanation work, thereby preserving the existing distinction unscathed.

For example, if an observed result does not match the assumptions concerning maximum operational capacity, repeatedly adding explanations such as "a different condition was actually in effect at that time" would make it possible to reconcile the concept of maximum operational capacity with any observed result whatsoever, without ever exposing it to falsification. If such additions are repeated, the concept becomes, in principle, unfalsifiable, and the premise of this paper—that the validity of a distinction is to be examined through observation—ceases to function.

Therefore, when an observed result does not match an existing conceptual distinction, priority is given to first making that mismatch explicit and considering the possibility that the distinction itself requires revision. Introducing auxiliary concepts or additional conditions solely to resolve a mismatch, without independent observational grounds, and thereby securing consistency on an ad hoc basis, is incompatible with the position of this paper.

This point stands in tandem with the provisional nature of the conceptual distinctions presented in this chapter. An open stance—that the distinctions may be revised through future observation—and a constraint—that such revision is not to be evaded through post-hoc auxiliary concepts—together are what sustain this paper's conceptual organization as falsifiable.

Observer Bias

Whereas the risk discussed in the previous section concerned the preservation of the theoretical structure itself, the limitation considered here concerns the observer who performs the inference.

In SCOPE, inference is carried out using outputs that appear in Malkuth and the situations related to those outputs as clues. However, every inference is made by some observer, and the observer's existing hypotheses, expectations, and patterns of attention may influence what is noticed and how the inference is formed. If a particular image of a person has already been assumed, information consistent with that image may receive greater attention, while inconsistent information may be overlooked or given less weight. Likewise, once a particular explanation has been adopted, subsequent observations may be more readily interpreted in ways that confirm that explanation.

This is not limited to human observers. Inferences made by LLMs may also become more likely to favor particular explanations in ways shaped by conversation history, recently presented hypotheses, or perceived expectations. Observer bias therefore cannot be treated as a problem specific to either humans or LLMs alone.

Observer bias is not necessarily eliminated simply by changing observers or by using multiple observers. The possibilities and limitations of using multiple observers are addressed in the next section. The point to establish here is that observer bias is a general limitation that may accompany the inferential procedures used in this paper and cannot necessarily be completely removed by any particular operational method.

Risk of Narrative Generation in Iterative Deepening

As discussed in Chapter 4, iterative SCOPE deepening carries a risk of narrative generation. When observable information is insufficient, repeatedly asking "why" may lead to the addition of causal relationships that have not actually been observed. To reduce this risk, at each point in the iterative process it is necessary to distinguish between information that was actually observed, inferences derived from the preceding analysis, and hypotheses that remain under consideration. If no additional information is available, the process should stop

with the status of unobservable, indeterminate, or with multiple hypotheses remaining on hold. The details are as described in Chapter 4.

The Same Method May Not Apply to Humans and LLMs

This paper treats both humans and LLMs as potential targets of observation. However, this does not mean that both can necessarily be estimated in the same way using identical observational methods or criteria. Humans and LLMs may differ in the processes by which outputs are generated, the range of information they can reference, and the factors that may influence their responses. Therefore, a method that is valid for one may not be directly applicable to the other.

At the same time, combining humans and LLMs as multiple observers may make it possible to detect some of each other's biases. However, this does not mean that observer bias is eliminated. LLM outputs may also be influenced by factors such as conversation history, hypotheses presented immediately beforehand, or inferred expectations. Therefore, neither humans nor LLMs can simply be treated as correction devices for the other.

Accordingly, even when humans and LLMs are used together, it is necessary to consider differences in their observational and estimation conditions and to examine individually under which conditions each observational method may be valid.

External Validation, Reproducibility, and Falsifiability

The conceptual distinctions and provisional organization presented in this paper have not been established as valid merely by being organized within this paper. Whether these distinctions are valid must ultimately be examined through verification by third parties other than the author, and their validity cannot be guaranteed by this paper alone.

For external validation to be conducted in a comparable manner, a certain degree of reproducibility is necessary. That is, under the same or similar conditions, observers other than the author should be able to identify similar tendencies or reach comparable judgments. However, as discussed in Chapter 5, concrete procedures for observing and measuring the dynamic characteristics addressed in this paper have not yet been established. Therefore, the conditions required for comparable validation are not yet sufficiently developed.

In addition, the conceptual distinctions presented here must, at minimum, allow the conditions under which they should be revised or rejected to be specified. Because the observational and measurement methods themselves remain undeveloped, such conditions cannot yet be presented in a rigorous form. However, as stated in the preceding sections, when observations do not correspond with a proposed distinction, the distinction itself should be reconsidered, and such reconsideration should not be avoided by introducing post hoc auxiliary concepts. This serves as an operational principle intended to prevent the framework from becoming insulated from falsification.

Accordingly, whether the conceptual distinctions proposed in this paper are academically valid should be judged through future independent examination by researchers other than

the author, attempts at replication, and efforts to falsify them. This paper is limited to presenting a starting point for such examination.

The Entire Paper Is a Provisional Hypothesis

As discussed above, the boundaries between the conceptual distinctions presented in this paper remain provisional, and procedures for observation and measurement have not yet been established. Observer bias has not been eliminated, there is no guarantee that the same methods can be directly applied to both humans and LLMs, and external validation and confirmation of reproducibility remain tasks for future work.

Given these limitations, this paper should not be regarded as presenting a completed theory of SCOPE's dynamic characteristics. Rather, it is an attempt to articulate, in a provisionally distinguishable form, phenomena that had previously been compressed into a single expression. The distinctions themselves are not intended as final answers. Their purpose is to make it possible to revisit, in future work, what kinds of observations may be used to examine each distinction.

Accordingly, the significance of this paper lies not in establishing a definitive classification of dynamic characteristics, but in distinguishing and presenting the objects of inquiry for future observation, validation, and attempts at falsification.

The Problem of Uniquely Determining Structure from Observation

When inferring the underlying structure that produced a single output observed in Malkuth, there is not necessarily only one pathway that could have led to that output. Even the same output may arise from different combinations of layers or from different states of dynamic characteristics. Therefore, it is not possible to uniquely determine, from the observed output alone, which structure produced it.

This is related to the issue discussed in Chapter 5, in which processes corresponding to multiple layers may be involved in a single task, but the problem addressed here is broader. Even if the observational task and observational conditions are carefully arranged, there is no principled guarantee of a one-to-one correspondence between an observable output and the internal structure that produced it. Multiple different structures may still produce the same or very similar outputs.

Accordingly, a structure inferred from an output should be treated as one candidate that is consistent with that output, rather than being established as the single correct explanation. When necessary, multiple candidates should be retained, leaving open the possibility of narrowing them within the range of information available at that time. This point also connects with the concept of inter-hypothesis parallel holding discussed in Chapter 3.

Conclusion

This paper has provisionally organized several concepts that can currently be distinguished among the dynamic characteristics of SCOPE that had previously been compressed into expressions such as "activation level." It distinguished layer activation level, maximum activation level, and main layer, further separated inter-hypothesis parallel holding from parallel freedom, and also organized the concept of iterative SCOPE deepening.

These distinctions are not intended to rank the capabilities of a target or to classify it into a fixed personality type. Rather, they are intended to support examination without conflating multiple possible cognitive structures or dynamic characteristics that may underlie the same observable output.

At the same time, the distinctions presented in this paper are not final classifications, and methods for observation and measurement have not yet been established. Future observation and examination may show that some currently separate concepts should be treated as one, that some should be divided further, that some should be redefined, or that some should no longer be retained.

SCOPE is not a coordinate system for fixing a target into a single interpretation based on one observable output. The dynamic characteristics organized in this paper are likewise positioned as provisional clues for distinguishing and examining multiple cognitive structures that may underlie that output.